

Integrated Multivariate Analysis of TCGA PanCanAtlas Transcriptomic Data

STAT 662 Final Presentation

Study question

Can high-dimensional pan-cancer RNA-seq profiles be compressed into stable latent structure that is clinically meaningful, predictive of cancer type, and informative for unsupervised tumor grouping?

10,055

Representative tumor
samples

32

Cancer types

0.893

Best held-out macro-F1

0.541

Best clustering ARI

Research Problem

Why this is a challenging statistical problem

- ▶ Each sample starts with 20,531 gene-expression variables.
- ▶ Gene measurements are highly correlated and biologically structured.
- ▶ The cohort spans 32 cancer types with uneven class sizes.
- ▶ Some cancers have overlapping transcriptomic programs, making prediction and clustering nontrivial.

One integrated pipeline links molecular structure, clinical association, prediction, and unsupervised recovery from the same transcriptomic signal.

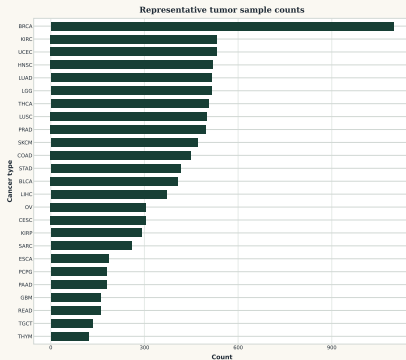
Study objectives

1. Summarize dominant transcriptomic variation with PCA.
2. Quantify how age, sex, stage, and cancer type explain latent expression scores.
3. Predict cancer type from a shared low-dimensional expression representation.
4. Test whether unlabeled tumors recover known structure through clustering.

Data and Cohort Construction

Cohort design

- ▶ Parse each TCGA barcode to recover patient identity and sample type.
- ▶ Keep tumor-derived specimens only.
- ▶ Select one **representative sample per patient**, prioritizing primary solid tumor.
- ▶ Harmonize with clinical covariates and clean the expression matrix.

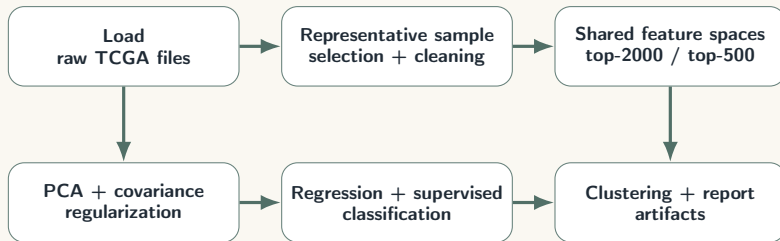


Largest cohorts:

BRCA 1,097; KIRC 533; UCEC 532; HNSC 520; LUAD 515.

11,069 manifest rows | 10,274 patients | 10,055 final cohort | 8,044 / 2,011 train / test.

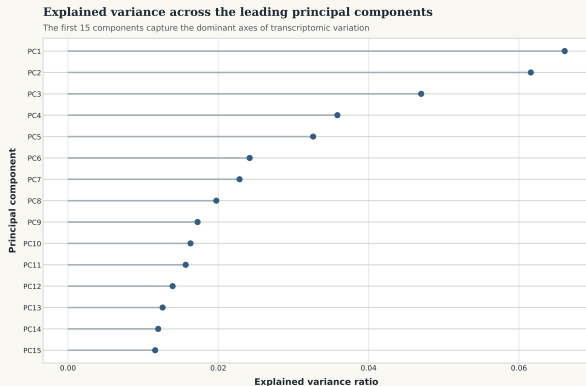
Analysis Workflow



Evaluation settings

Classification uses a stratified 80/20 split with 5-fold CV; PCA stability uses 10 repeated 80% subsamples; clustering stability is measured by repeated-subsample ARI.

PCA and Covariance Results



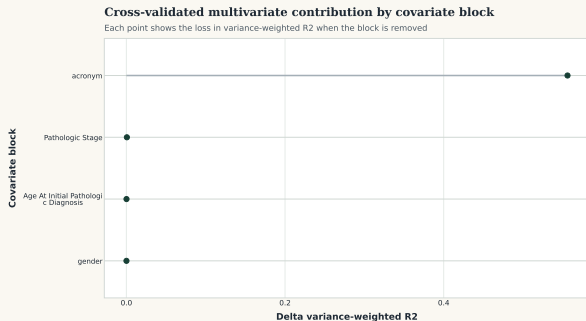
Dimensionality

- ▶ PC1 variance: 6.61%
- ▶ First 10 PCs: 34.35%
- ▶ First 50 PCs: 57.94%

Stability and covariance

- ▶ PCs 1–5 loading correlation: > 0.997
- ▶ Mean subspace similarity: 0.9995
- ▶ Graphical Lasso penalty: 0.5543

Multivariate Regression on Latent Expression Scores



Key regression findings

- ▶ Response: first 10 PC scores; predictors: age, sex, stage, and cancer acronym.
- ▶ Multivariate ridge with 5-fold CV gives variance-weighted $R^2 = 0.6944$.
- ▶ Overall permutation significance remains detectable: $p = 0.0385$.
- ▶ Cancer acronym is dominant (delta- $R^2 = 0.5561$); stage is modest; age and sex are negligible after conditioning on cancer type.

Supervised Cancer-Type Prediction: Experimental Setup

Representation

- ▶ Shared top-2,000 gene matrix
- ▶ Train-only standardization
- ▶ PCA embedding with 50 PCs

Models compared

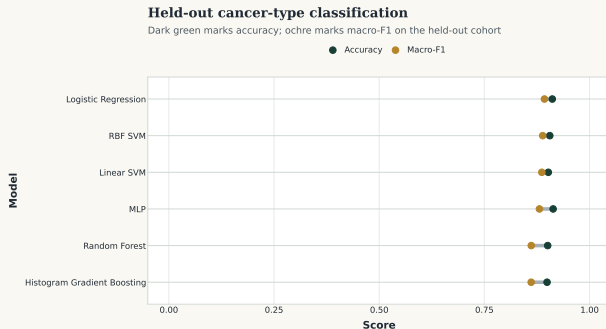
Logistic Regression, Linear SVM, RBF SVM, Random Forest, Histogram Gradient Boosting, and Multilayer Perceptron.

Validation protocol

- ▶ Stratified 80/20 split
- ▶ Training rows: 8,044
- ▶ Test rows: 2,011
- ▶ Hyperparameter tuning: 5-fold CV
- ▶ Selection metric: macro-F1

Macro-F1 emphasizes balanced performance across all 32 cancers rather than letting the largest classes dominate the score.

Supervised Classification Results



Best overall model

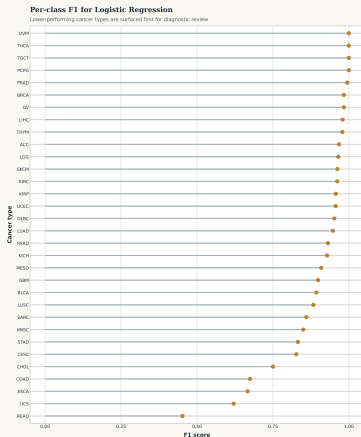
Logistic Regression

- ▶ Accuracy: **0.9115**
- ▶ Macro-F1: **0.8928**
- ▶ Weighted F1: **0.9141**
- ▶ Best CV macro-F1: **0.8727**

Important comparison

MLP has slightly higher raw accuracy (**0.9135**) but lower macro-F1 (**0.8809**), so Logistic Regression is the strongest balanced model.

Classification Diagnostics



Class-level performance

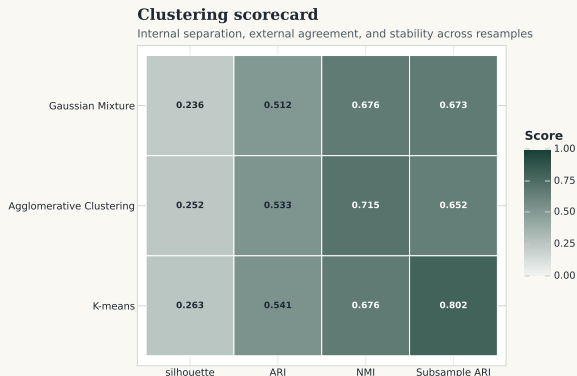
- ▶ 20 of 32 cancer types achieve F1 ≥ 0.90
- ▶ Near-perfect: UVM, THCA, TGCT, PCPG, PRAD, BRCA
- ▶ Hardest: READ, ESCA, COAD, UCS

Most important confusion pairs

Confusion	Count
COAD $\rightarrow$ READ	32
READ $\rightarrow$ COAD	12

Dominant errors occur in biologically adjacent disease groups rather than arbitrary label swaps.

Unsupervised Clustering Results



Setup

- ▶ Shared top-2,000 genes
- ▶ PCA to 20 PCs
- ▶ Number of clusters fixed at 32
- ▶ K-means, agglomerative, Gaussian mixture

Best results

K-means

- ▶ Silhouette: 0.2630
- ▶ ARI: 0.5406
- ▶ NMI: 0.6762

K-means also gives the strongest stability, with subsample ARI 0.8020; agglomerative clustering has the highest NMI at 0.7149.

Integrated Interpretation

0.9995

Mean PCA subspace
similarity

0.6944

Regression CV R^2

0.8928

Best macro-F1

0.8020

Best clustering stability

What the full study shows

- ▶ The transcriptome has strong low-dimensional organization and the leading PCs are highly reproducible.
- ▶ Cancer type is the dominant driver of latent expression variation, with stage contributing a smaller secondary effect.
- ▶ Supervised prediction is stronger than unsupervised recovery, but both analyses reveal meaningful tumor structure.

Limitations and Next Steps

Current limitations

- ▶ Expression-only analysis; no mutation, methylation, or copy-number integration
- ▶ One representative tumor sample per patient
- ▶ Clustering was evaluated at a fixed $K = 32$

Natural next steps

- ▶ Add multimodal genomic data
- ▶ Move from cancer-type prediction to subtype discovery
- ▶ Validate the learned representation on an external cohort

Reproducibility: `run_tcga_pipeline.py` executes the full workflow and writes all report figures to `artifacts/tcga/`.

Thank you

Questions?

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